

Bible Expert Project Agent — Enhancement Report

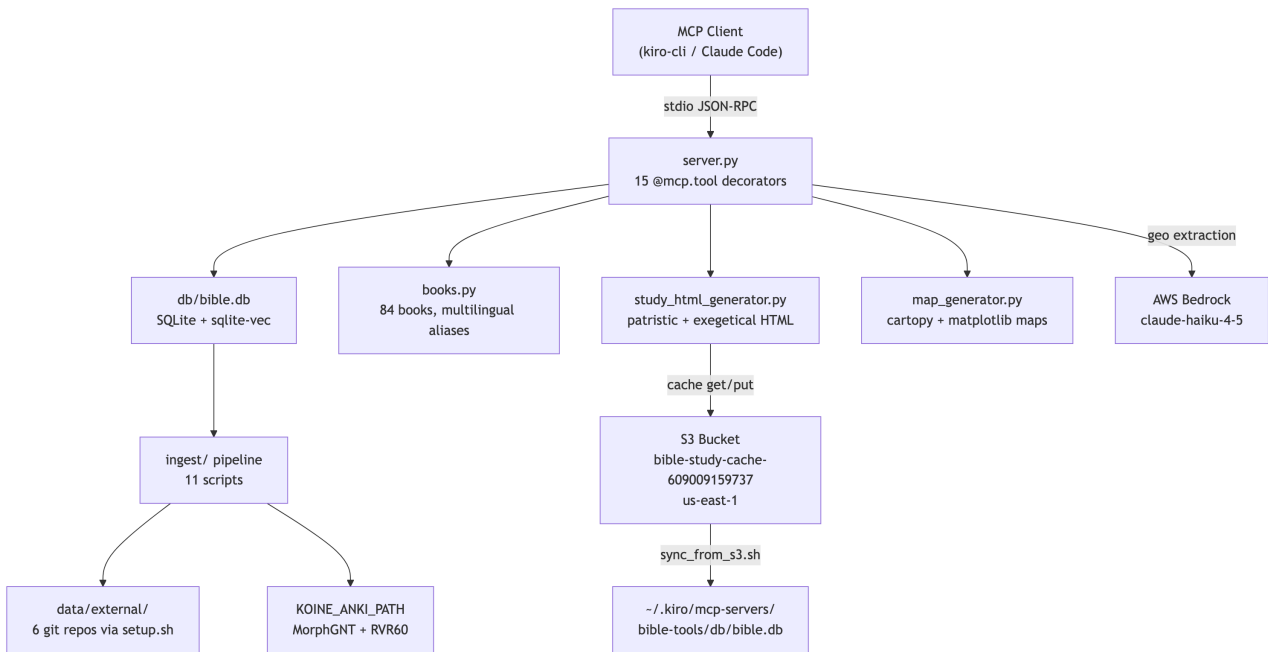
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Executive Summary

The `bible-expert` project is a **Python 3.11+ MCP server** (`server.py`, ~988 lines) exposing **15 biblical research tools** over stdio transport via FastMCP. It queries a SQLite database (`db/bible.db`) with 7 text versions spanning Greek, Hebrew, Latin, Spanish, and English, plus patristic commentary, Dead Sea Scrolls, and semantic embeddings.

The current agent prompt (829 chars) is a generic template missing all project-specific context. Two active bugs were confirmed: `verse_lookup` defaults to `"SBLGNT"` (not a valid DB version — Greek NT is stored as `"MorphGNT"`), and both README and `pyproject.toml` report 12 tools when the actual count is 15. The enhanced prompt below corrects both and gives any agent working on this project immediate productivity.

Architecture Diagram



Detailed Analysis

Project Structure

Path	Role
server.py	Entry point — all 15 MCP tools, ~988 lines
books.py	Canonical book resolution, 84 books (IDs 1–84), multilingual aliases
bible_places.py	Geographic coordinates for map generation
map_generator.py	matplotlib + cartopy map PNG generation
study_html_generator.py	Interactive HTML output with S3 cache
ingest/	11 data pipeline scripts
data-raw/	Local raw data (RVR60 JSON, UBS5 apparatus, Matthew HTML notes)
db/bible.db	SQLite DB — gitignored, populated by ingest or sync_from_s3.sh
Makefile	All developer workflows
setup.sh	Downloads 6 external git repos + runs ingest

15 MCP Tools

Tool	Purpose
book_list	List all 84 supported books
verse_lookup	Fetch verses by book/chapter/verse, any version
parallel_versions	Same passage across multiple versions side-by-side
semantic_search	Embedding-based similarity search
morphology_analysis	Greek/Hebrew morphology with Strong's numbers
critical_apparatus	Textual variants from UBS5/NA28 apparatus
patristic_commentary	Church father commentary on a passage
save_patristic_original	Persist patristic text to DB
cross_references	Cross-reference lookup
word_study	Deep lexical study for a word/lemma
authenticity_report	Textual authenticity analysis
dss_lookup	Dead Sea Scrolls parallels

Tool	Purpose
canon_history	Canonical status and history of a book
text_comparison	Diff two versions of a passage
chapter_study	Full chapter study with Bedrock geo extraction

Text Versions (DB version strings)

Version String	Content
MorphGNT	Greek New Testament (morphologically tagged)
LXX	Septuagint (Greek OT)
WLC	Westminster Leningrad Codex (Hebrew OT)
RVR60	Reina-Valera 1960 (Spanish)
YLT	Young's Literal Translation (English)
Vulgate	Latin Vulgate
ApostolicFathers	Apostolic Fathers corpus

DB Schema (Key Tables)

```
verses(book, chapter, verse_num, version, text, testament, canon_status, morphology)
morphology(book, chapter, verse_num, version, word_pos, word, lemma, morph_code, gloss, strongs)
apparatus(book, chapter, verse_num, variant_id, reading, manuscripts, text_type)
patristic(book, chapter, verse_num, father, work, text, date_approx, source_collection)
cross_refs, dss, verse_embeddings
```

AWS Resources

Resource	Value
S3 Bucket	bible-study-cache-609009159737 (us-east-1)
Bedrock Model	global.anthropic.claude-haiku-4-5-20251001-v1:0
Region	us-east-1

Bugs Found

Bug 1 — `verse_lookup` wrong default version (Active)

`verse_lookup` defaults to `version="SBLGNT"` but `SBLGNT` is not a valid DB version string. Greek NT is stored as `MorphGNT`. Any call with the default returns empty results.

Fix (1 line in `server.py`):

```
# Change line ~49 from:  
def verse_lookup(..., version: str = "SBLGNT", ...  
# To:  
def verse_lookup(..., version: str = "MorphGNT", ...
```

Bug 2 — Outdated tool count in docs

README and `pyproject.toml` both say "12 tools". Actual count is **15**. Missing from docs: `book_list`, `save_patristic_original`, `chapter_study`.

Fix:

```
# In README.md and pyproject.toml description, update "12 tools" → "15 tools"  
# Add to README tools table: book_list, save_patristic_original, chapter_study
```

Common Workflows

```
# Initial setup
make setup
./setup.sh

# Start MCP server
make run

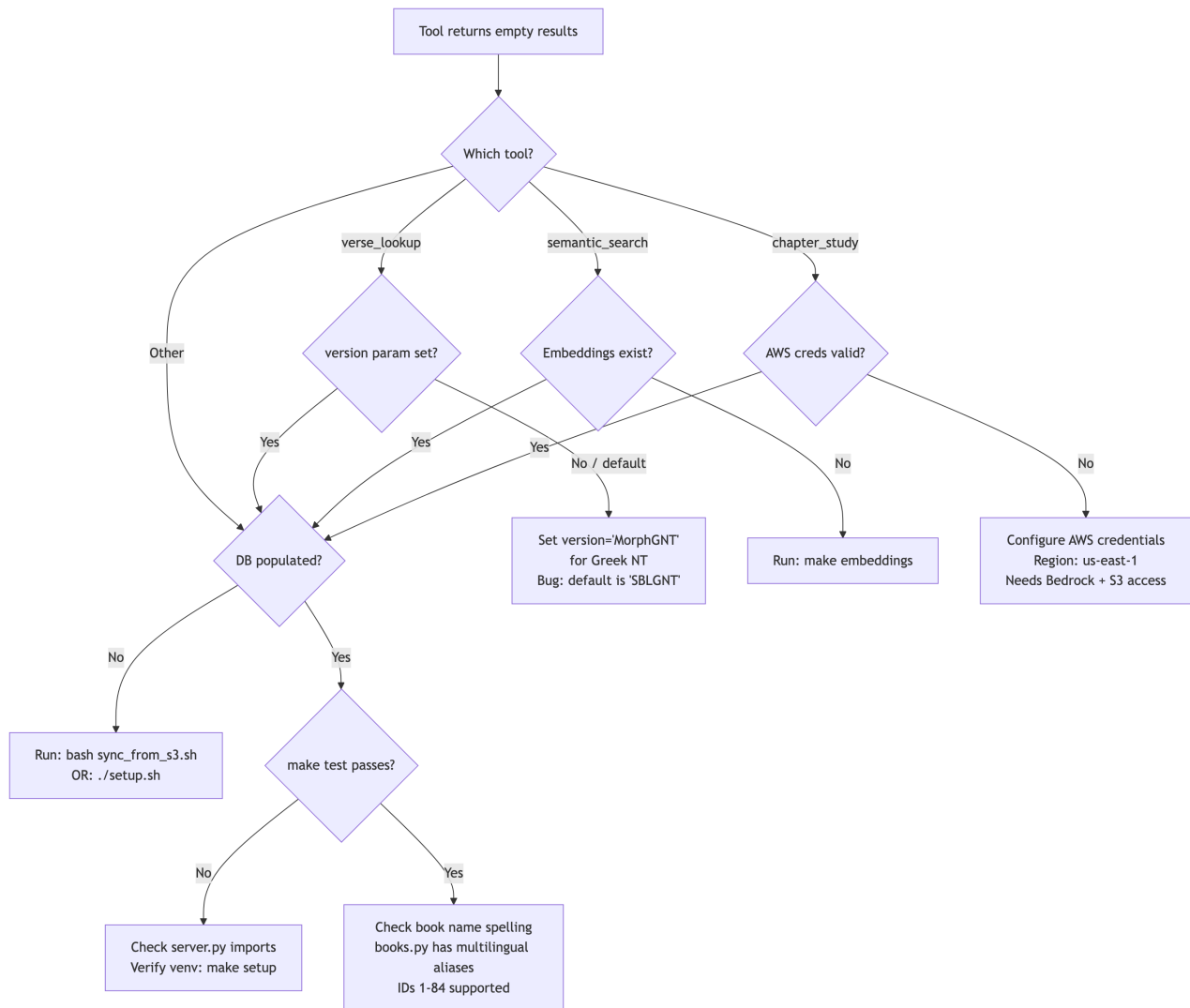
# Smoke test (verifies tool count)
make test

# Sync DB from S3
bash sync_from_s3.sh

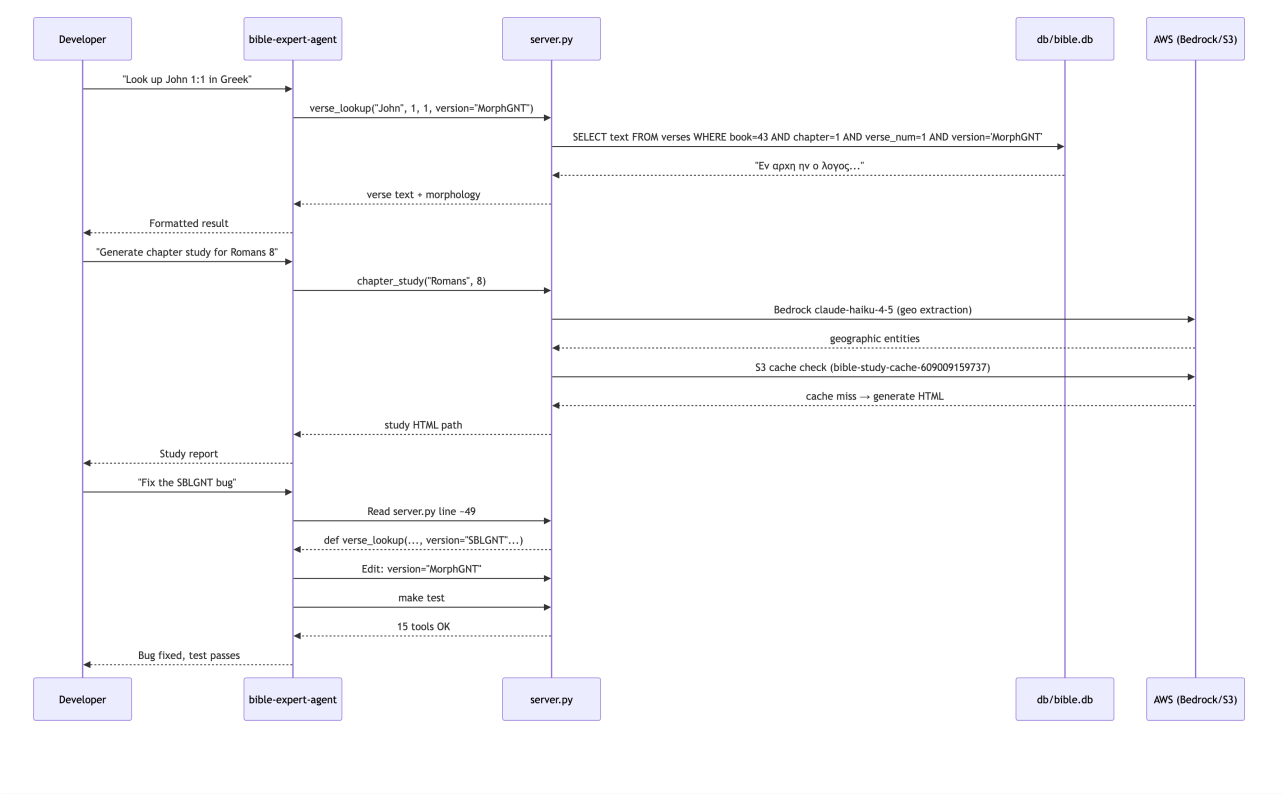
# Regenerate embeddings
make embeddings

# Ingest from local Koine Anki repo
KOINE_ANKI_PATH=/path/to/koine-anki make ingest-local
```

Troubleshooting Decision Tree



Agent Enhancement Sequence



Action Plan

Priority	Action	Effort	Impact
1	Deploy enhanced agent prompt (JSON below)	2 min	High — immediate agent productivity
2	Fix <code>verse_lookup</code> SBLGNT default → MorphGNT	1 line	High — fixes silent data bug
3	Update README + pyproject.toml tool count 12→15	5 min	Medium — doc accuracy
4	Add minimal <code>tests/</code> directory	30 min	Medium — catch regressions

Enhanced Agent JSON

```
{
  "name": "bible-expert-project-agent",
  "description":
    "Specialized agent for bible-expert – Python MCP server with 15 biblical research tools (verse lookup, morphology, patristic commentary, DSS, semantic search, chapter study)",
  "prompt": "You are a specialized developer for the bible-expert project.
\n\n## Project\n- Path: ~/work/github/bible-expert/\n- Purpose: Python 3.11+ MCP server (FastMCP, stdio transport) with 15 biblical research tools\n- Entry point: server.py (~988 lines, 15 @mcp.tool decorators)\n- DB: SQLite at db/bible.db (gitignored – populated by ingest pipeline or sync_from_s3.sh)\n\n## Architecture\n- server.py – all 15 MCP tools + helper functions\n- books.py – canonical book resolution (84 books, IDs 1–84, multilingual aliases)\n- bible_places.py – geographic coordinates for map generation\n- map_generator.py – matplotlib+cartopy map PNG generation\n- study_html_generator.py – interactive HTML with S3 cache (bucket: bible-study-cache-609009159737, us-east-1)\n- ingest/ – 11 data pipeline scripts\n- data-raw/ – local raw data (RVR60 JSON, UBS5 apparatus, Matthew HTML notes)\n\n## 15 Tools\nbook_list, verse_lookup, parallel_versions, semantic_search, morphology_analysis, critical_apparatus, patristic_commentary, save_patristic_original, cross_references, word_study, authenticity_report, dss_lookup, canon_history, text_comparison, chapter_study\n\n## Text Versions (DB version strings)\nMorphGNT (Greek NT), LXX (Septuagint), WLC (Hebrew OT), RVR60 (Spanish 1960), YLT (English), Vulgate (Latin), ApostolicFathers\n\n## DB Schema (key tables)\nverses(book, chapter, verse_num, version, text, testament, canon_status, morphology)\nmorphology(book, chapter, verse_num, version, word_pos, word, lemma, morph_code, gloss, strongs)\napparatus(book, chapter, verse_num, variant_id, reading, manuscripts, text_type)\npatristic(book, chapter, verse_num, father, work, text, date_approx, source_collection)\ncross_refs, dss, verse_embeddings\n\n## Common Workflows\n``bash\nmake setup\n./setup.sh\nmake run\nmake test\nmake ingest-local\nmake embeddings\nbash sync_from_s3.sh\n```\n\n## Critical Rules\n- ALWAYS use version string 'MorphGNT' (not 'SBLGNT') for Greek NT – 'SBLGNT' is not a valid DB version and returns empty results\n- Book IDs 1–66 = Protestant canon; 67–84 = deuterocanonical/extra-biblical\n- db/bible.db is gitignored – never commit it; use sync_from_s3.sh or run ingest pipeline\n- AWS credentials required for: chapter_study (Bedrock claude-haiku-4-5) and study_html_generator.py (S3 cache)\n- Run make test after any server.py change to verify tool count stays at 15\n- No ARCHITECTURE.md, DEVELOPMENT.md, or TROUBLESHOOTING.md – consult README.md and server.py\n- docstrings\n- No tests/ directory – make test is the only automated check\n- KOINE_ANKI_PATH env var must point to koine-anki repo for ingest/ingest_local.py",
  "tools": ["read", "write", "glob", "grep", "code", "shell", "@kiro-checkpoint/*"],
  "allowedTools": ["read", "write", "glob", "grep", "code", "shell", "web_search", "web_fetch", "checkpoint", "diff", "rollback", "list_checkpoints", "init", "branch", "switch_branch"],
  "includeMcpJson": false
}
```


Key Findings Summary

Finding	Status	Detail
Tool count	15 (not 12)	<code>grep @mcp.tool server.py</code> = 15 hits
Language/Framework	Python 3.11+ / FastMCP	<code>pyproject.toml</code> , <code>mcp>=1.27.0</code>
DB	SQLite + sqllite-vec	<code>db/bible.db</code> , gitignored
Text versions	7	MorphGNT, LXX, WLC, RVR60, YLT, Vulgate, ApostolicFathers
Book coverage	84 books	IDs 1–84: Protestant + deuterocanonical + pseudepigrapha
AWS services	Bedrock + S3	claude-haiku-4-5, bucket <code>bible-study-cache-609009159737</code>
Bug: SBLGNT default	Active	<code>verse_lookup</code> default returns empty — fix: <code>"MorphGNT"</code>
Bug: doc tool count	Active	README + pyproject say 12, actual is 15
Test suite	None	<code>make test</code> smoke test only; no <code>tests/</code> directory
Missing docs	Confirmed	No ARCHITECTURE.md, DEVELOPMENT.md, TROUBLESHOOTING.md

References

Source	Notes
<code>server.py</code>	15 tools, all signatures, Bedrock model ID (line 972)
<code>pyproject.toml</code>	Dependencies, Python version, outdated tool count
<code>Makefile</code>	All make targets
<code>README.md</code>	Project overview — tool count outdated (says 12)
<code>ingest/init_schema.py</code>	Full DB schema
<code>books.py</code>	84-book canonical resolution
<code>study_html_generator.py</code>	S3 bucket name and caching logic

Source	Notes
<code>sync_from_s3.sh</code>	S3 sync target path
<code>setup.sh</code>	External data sources and ingest sequence